

Daniel W Liu

Computer Science and Finance Student

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, C, HTML/CSS, Racket

Frameworks: React.js, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, Bootstrap

Cloud & Databases: AWS, Google Cloud, PostgreSQL, MongoDB, Redis, Supabase

Data & Libraries: pandas, NumPy, OpenCV, Matplotlib, yFinance, PyQt5, Three.js

DevOps & Tools: Docker, Git, GitHub, Linux, Vercel, Canva, Unity, Blender

EXPERIENCE

Software Engineering Intern

Jan 2026 – Present

Wedge YC S25 Healthcare AI Startup

San Francisco, CA (Remote)

- Designed an AI-powered voice agent system for insurance claim follow-ups, integrating LLMs, Vapi, and Deepgram STT to autonomously navigate IVR workflows and extract structured claim data across **50+ clinical sites**.
- Developed a production claims dashboard using Next.js and Supabase, enabling AI-driven CSV ingestion, persistent claim state across **45+ structured data fields**, and AI data extraction from transcripts through webhooks.

Fullstack and Events Intern

Sept 2025 – Dec 2025

HOSA Canada

Toronto, ON

- Engineered a robust full-stack educational resource sharing platform using Next.js, AWS, and MongoDB, serving **9,000+ Canadian students** competing across **50+ unique associations** spanning North America and Asia.
- Scaled an automated SMS communications pipeline using Express.js, Twilio, AWS, and Redis, leveraging queues, middleware rate limiting, and unit-tested workflows to coordinate logistics for **360+ professional workshops**.
- Implemented multi-role authorization with NextAuth.js and protected API routes, enforcing role-based access control across **35+ competitive events** with integrated AI content moderation for safe user submissions.

Computer Vision and ML Research Intern

Jul 2025 – Aug 2025

Pearl Lab Holland Bloorview Kids Rehabilitation Hospital

Toronto, ON

- Developed real-time AI object detection systems using OpenCV and pupil-apriltags for *Bootle Blast*, a gamified rehabilitation platform clinically tested at **3+ Canadian hospitals** for children's motor skill rehabilitation.
- Analyzed and benchmarked ML object detection models using NumPy and pandas against YOLOv8 on a dataset of **250+ unique environmental conditions**, improving **model confidence by 100%** in real-world scenarios.

PROJECTS

rainbolt.ai | *Next.js, Three.js, FastAPI, OpenAI CLIP, Pinecone, LangChain, Google Cloud*

Devpost GitHub

- Awarded **Best UI Hack** at UofT Hack the Valley X as **frontend lead**, delivering an AI interface with Next.js, Three.js, and Zustand for interactive dashboards and 3D visualizations, deployed with Vercel and Google Cloud.
- Architected a RAG pipeline with a dataset of **955,199+ images**, integrating OpenAI CLIP, Pinecone, LangChain, and Mapillary APIs with a FastAPI backend for accurate location inference and contextual response generation.

VibeTrade | *Next.js, FastAPI, Supabase, LangGraph, Alpaca, Finnhub, ElevenLabs*

Devpost GitHub

- Built an agentic AI trading dashboard that analyzes market data, news, and portfolio context to generate real-time actionable insights and explainable trading rationales, integrating **5 financial APIs** over live WebSocket streams.
- Implemented a customizable and animated 3D AI voice agent with STT/TTS, background workers for market monitoring, and a FastAPI + LangGraph workflow enabling voice-triggered trade execution and reasoning.

Curve Guard | *React.js, Express.js, Node.js, AWS, MongoDB, Three.js, MediaPipe*

GitHub

- Orchestrated a MERN posture monitoring system with real-time MediaPipe pose detection (**33 landmarks**), RESTful APIs, and MongoDB-backed analytics, deployed with AWS and Express.js, achieving **95%** accuracy.
- Constructed a responsive React.js and Three.js frontend with dynamic visualization and alerts, reducing **posture errors by 80%** through real-time correction across 3 key metrics: head tilt, shoulder unevenness, and slouch.

EDUCATION

University of Waterloo | *Double Major in Computer Science and Finance*

Sept 2025 – Present

Bachelor of Computing and Financial Management (Co-op) — GPA: 3.98/4.0

Waterloo, ON

International Baccalaureate Diploma Programme

Sept 2021 – May 2025

IB Diploma — 45/45 Perfect Score — Top 0.5% Internationally

Thornhill, ON